FARMHUB

A central hub for farming products and equipment

ANDROID MOBILE APPLICATION

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PROJECT OVERVIEW

Introduction

FarmHub is an Android platform that brings agricultural products and equipment rental together in one place. It connects customers with agricultural sellers and equipment owners, while providing role-based access and Firebase-backed data management.

Project focus

Make essential farming resources easier to discover, access and coordinate.

01 One platform

Browse farm inputs and request equipment rentals without switching between separate services.

02 Shared resources

Equipment owners can list machinery that may otherwise remain unused.

03 Digital workflow

Authentication, orders, bookings and inspection records are managed in the app.

Problem Statement & Existing System

! High equipment cost

Many farmers cannot justify purchasing machinery that may be needed only during specific seasons.

↔ Underused machinery

Privately owned equipment can remain idle while nearby farmers may need the same resource.

+ Scattered product access

Seeds, fertilizers and crop-care products are often sourced from different sellers and channels.

Opportunity: connect agricultural product discovery and farmer-to-farmer equipment rental in a single mobile application.

WHAT THE APPLICATION AIMS TO ACHIEVE

Objectives of FarmHub

01 Centralize access

Bring agricultural products and equipment discovery into one mobile platform.

02 Enable equipment sharing

Help customers request rentals from approved equipment owners.

03 Supports Cash on Delivery

Manage cart items and place product orders through Cash on Delivery.

04 Organize workflows

Track rental requests, owner decisions, agreement acceptance and return inspection.

05 Control access by role

Provide separate customer, owner and admin navigation.

HOW FARMHUB ADDRESSES THE PROBLEM

Proposed Solution

CUSTOMERS

Discover products
and request rentals

OWNERS / SELLERS

List equipment
and farm products

FARMHUB

One mobile experience for products + rentals

FIREBASE

Authentication
and cloud data

ADMIN

Review owner applications
and view commission summary

ROLE-BASED NAVIGATION

User Roles

01 Customer

- Register / sign in
- Browse products
- Manage cart and COD orders
- Browse equipment
- Submit rental requests
- View own orders and bookings

02 Equipment Owner / Seller

- Apply for owner access
- Add and edit equipment listings
- Manage product listings
- Review rental requests
- Approve or reject bookings
- Record return inspection

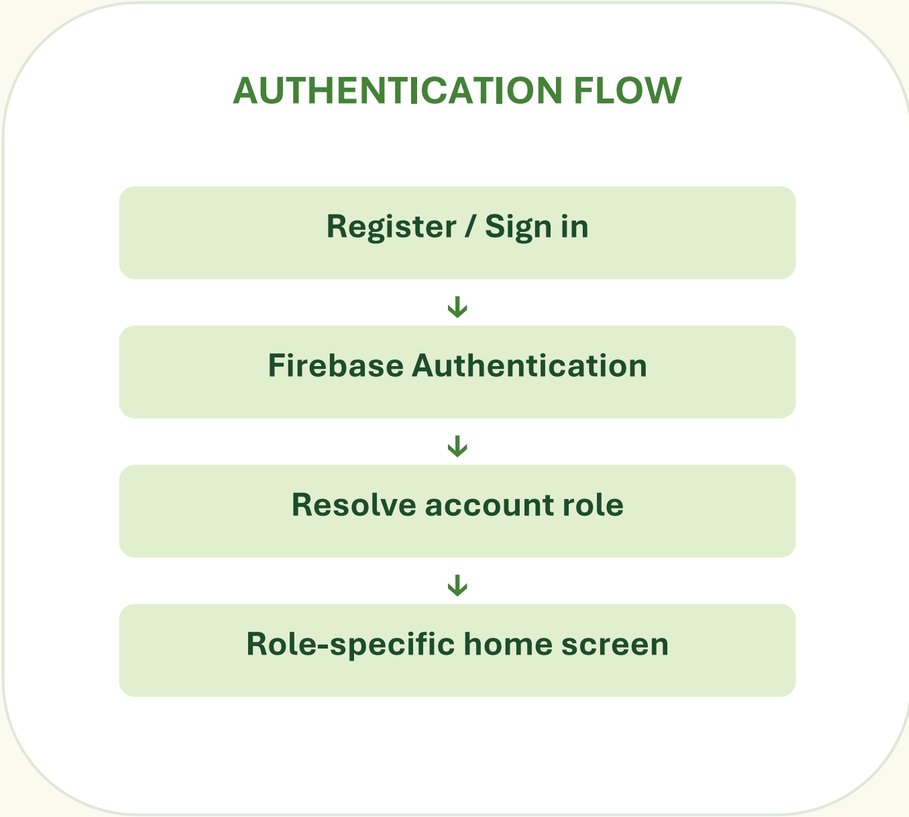
03 Administrator

- Review pending owner applications
- Approve or reject applications
- View revenue summary
- Access protected admin screens

SECURE SIGN-IN AND ROLE RESOLUTION

User Authentication

- Email and password registration and login using Firebase Authentication.
- Sign-out functionality for active sessions.
- Registration supports Customer or Owner application selection.
- Owner registration creates a pending application for admin review.
- Role checks determine which areas of the app a signed-in user can access.



DISCOVER FARMING ESSENTIALS

Agricultural Product Marketplace

- Browse published agricultural products.
- View product details, price, category and seller information.
- Check listed stock availability.
- Search and filter products by category.
- Owners / sellers can add and manage their product listings.







Seeds • Inputs • Irrigation

SIMPLE CHECKOUT WITH COD

Shopping Cart & Product Orders

01 1. Add to cart

Select products and quantities from the marketplace. Cart items can be updated or removed.

02 2. Review order

Check selected items, quantities, total amount and required delivery address.

03 3. Place COD order

Submit the product order using Cash on Delivery. The order is saved in Firestore.

FIND AND REQUEST MACHINERY

Agricultural Equipment Rental

- Browse published equipment listings.
- View equipment details, rental price and location.
- Select a rental date and submit a booking request.
- Booking requests begin with Pending status.
- Customers can view the status of their rental requests.







Tractors • Tillers • Sprayers

TOOLS FOR APPROVED OWNERS

Owner & Seller Management

01 Equipment listings

Add equipment, edit listing details, manage availability and view owner equipment.

02 Booking management

Review incoming rental requests and approve or reject pending bookings.

03 Product management

Add agricultural products and manage the seller's existing product listings.

Admin Module

OWNER APPLICATION REVIEW

- Admin access is verified using an active admin record.
- Pending owner applications are listed for review.
- Admin can approve or reject an application.
- Approval creates an owner approval record used by role checks.

REVENUE SUMMARY

5%
Commission rate

Product orders + rental bookings

RECORD RENTAL TERMS AND EQUIPMENT CONDITION

Rental Agreement & Return Inspection

01 Agreement acceptance

The customer accepts the rental agreement terms as part of submitting the booking request.

02 Booking status

The request moves through Pending, Approved or Rejected based on the owner's decision.

03 Return inspection

The owner records return status, equipment condition, damage description and inspection charge.

Booking and inspection details are stored in Cloud Firestore.

ANDROID APPLICATION STACK

Tools & Technologies Used

Java

Main programming language for application logic.

XML

Layouts, screens and interface components.

Android SDK

Android platform APIs and build tools.

Firebase Authentication

Email/password sign-in and account management.

Cloud Firestore

Cloud database for profiles, products, orders, equipment and bookings.

Gradle

Build automation and dependency management.

ADB

Install and debug the app on a connected Android device.

Visual Studio Code

Source-code editing and project management.

Platform: Android | Architecture: Activity-based app + repository classes + Firebase backend

FROM MOBILE UI TO CLOUD DATA

System Architecture & Flow

ANDROID PRESENTATION — Activities • XML layouts • Role-based navigation

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APPLICATION LOGIC — Repositories • Product, equipment, booking and order workflows

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FIREBASE SERVICES — Firebase Authentication • Cloud Firestore

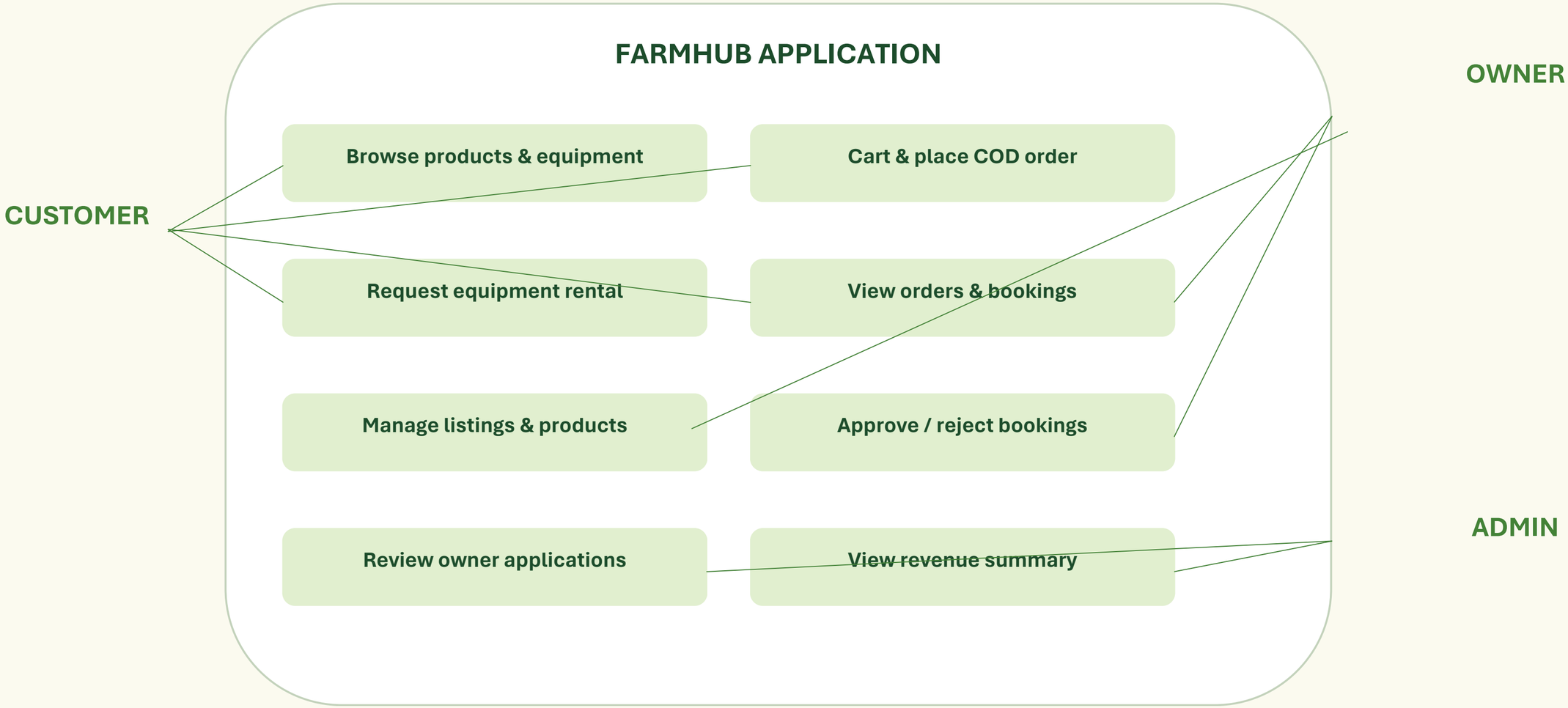
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DATA COLLECTIONS — userProfiles • products • equipment • orders • bookings • ownerApplications • admins

Firestore security rules help restrict access based on sign-in identity and approved roles.

MAIN INTERACTIONS BY USER ROLE

Use Case Overview



Conceptual use-case overview based on the implemented customer, owner and admin workflows.

Revenue Model & Future Enhancements

CURRENT REVENUE LOGIC

5%

Calculated commission rate

- Commission calculated on product orders.
- Commission calculated on equipment rentals.
- Admin screen summarizes recorded commission.

POSSIBLE FUTURE ENHANCEMENTS

- Integrate a secure online payment gateway.
- Expand admin tools and operational reports.
- Add order fulfilment and delivery tracking.
- Introduce richer notifications and booking reminders.
- Add analytics for product and equipment usage.

THANK YOU

Growing connections. Sharing resources.

